

1. Local Kafka cluster

docker ps -a command

```
> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
013e95ebb499   confluentinc/cp-kafka:7.4.0        "/etc/confluent/dock..." 8 minutes ago  Up 8 minutes  9092/tcp, 0.0.0.0:9093->9093/tcp, [...]9093->9093/tcp  homework-1-kafka2-1
5d5f7e96eb66   confluentinc/cp-kafka:7.4.0        "/etc/confluent/dock..." 8 minutes ago  Up 8 minutes  0.0.0.0:9092->9092/tcp, [...]9092->9092/tcp  homework-1-kafka1-1
f65ca76c8c8f   confluentinc/cp-zookeeper:7.4.0    "/etc/confluent/dock..." 8 minutes ago  Up 8 minutes  2181/tcp, 2888/tcp, 3888/tcp        homework-1-zookeeper-1
```

[CLI] creating a topic

```
> docker exec -it homework-1-kafka1-1 bash
[appuser@5d5f7e96eb66 ~]$ kafka-topics --create --topic my-kafka-topic --bootstrap-server localhost:29092 --partitions 2 --replication-factor 2
Created topic my-kafka-topic.
[appuser@5d5f7e96eb66 ~]$
```

[CLI] list topics

```
[appuser@5d5f7e96eb66 ~]$ kafka-topics --list --bootstrap-server localhost:29092
my-kafka-topic
[appuser@5d5f7e96eb66 ~]$
```

[CLI] send messages

```
[appuser@5d5f7e96eb66 ~]$ kafka-console-producer --topic my-kafka-topic --bootstrap-server localhost:29092
>Hello Kafka cluster!
>This is my second message.
>Testing the console producer.
>Sensor reading: temperature is normal.
>Processing transaction #001.
>Halfway there with the homework.
>Data streaming is working perfectly.
>Checking topic partitions.
>Message number nine.
>Final message, getting ready to exit.
[appuser@5d5f7e96eb66 ~]$
```

[CLI] receive messages

```
[appuser@5d5f7e96eb66 ~]$ kafka-console-producer --topic my-kafka-topic --bootstrap-server localhost:29092
>Hello Kafka cluster!
>This is my second message.
>Testing the console producer.
>Sensor reading: temperature is normal.
>Processing transaction #001.
>Halfway there with the homework.
>Data streaming is working perfectly.
>Checking topic partitions.
>Message number nine.
>Final message, getting ready to exit.
[appuser@5d5f7e96eb66 ~]$ kafka-console-consumer --topic my-kafka-topic --from-beginning --bootstrap-server localhost:29092
Hello Kafka cluster!
This is my second message.
Testing the console producer.
Sensor reading: temperature is normal.
Processing transaction #001.
Halfway there with the homework.
Data streaming is working perfectly.
Checking topic partitions.
Message number nine.
Final message, getting ready to exit.
^CProcessed a total of 10 messages
[appuser@5d5f7e96eb66 ~]$
```

[CLI] deleting a topic

```
[appuser@5d5f7e96eb66 ~]$ kafka-topics --delete --topic my-kafka-topic --bootstrap-server localhost:29092
[appuser@5d5f7e96eb66 ~]$
```

2. Local Redpanda cluster

docker ps -a

```
> docker ps -a
CONTAINER ID   IMAGE                                NAMES                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
fd9e8813a7cd   docker.redpanda.com/redpandadata/redpanda:latest   "/entrypoint.sh redp..." 4 seconds ago  Up 4 seconds  8081-8082/tcp, 9092/tcp, 0.0.0.0:9093->9093/tcp, [...]9093->9093/tcp, 0.0.0.0:9645->9644/tcp, [...]9645->9644/tcp  local-redpanda-cluster-redpanda-1-1
443d07c64748   docker.redpanda.com/redpandadata/redpanda:latest   "/entrypoint.sh redp..." 4 seconds ago  Up 4 seconds  0.0.0.0:9092->9092/tcp, [...]9092->9092/tcp, 8081-8082/tcp, 0.0.0.0:9644->9644/tcp  local-redpanda-cluster-redpanda-0-1
```

[CLI] creating a topic

```
redpanda@443d67c64748:/$ rpk topic create my-redpanda-topic -p 2 -r 1
TOPIC          STATUS
my-redpanda-topic  OK
redpanda@443d67c64748:/$
```

[CLI] list topics

```
redpanda@443d67c64748:/$ rpk topic list
NAME          PARTITIONS  REPLICAS
my-redpanda-topic  2          1
redpanda@443d67c64748:/$
```

[CLI] send messages

```
redpanda@443d67c64748:/$ rpk topic produce my-redpanda-topic
Hello Redpanda cluster!
Produced to partition 1 at offset 0 with timestamp 1783865941774.
Testing the rpk producer.
Produced to partition 1 at offset 1 with timestamp 1783865957493.
Raft consensus is working perfectly.
Produced to partition 1 at offset 2 with timestamp 1783865983162.
Message four of ten.
Produced to partition 1 at offset 3 with timestamp 1783865998596.
Stream processing is incredibly fast.
Produced to partition 1 at offset 4 with timestamp 1783865995456.
Sending data to our single replica.
Produced to partition 1 at offset 5 with timestamp 1783866001812.
Almost done with the homework.
Produced to partition 1 at offset 6 with timestamp 1783866006456.
Event-driven architecture rocks.
Produced to partition 1 at offset 7 with timestamp 1783866010460.
Message number nine.
Produced to partition 1 at offset 8 with timestamp 1783866013937.
Final Redpanda message, exiting now.
Produced to partition 1 at offset 9 with timestamp 1783866018615.
redpanda@443d67c64748:/$
```

[CLI] receive messages

```
redpanda@443d67c64748:/$ rpk topic consume my-redpanda-topic
{
  "topic": "my-redpanda-topic",
  "value": "Hello Redpanda cluster!",
  "timestamp": 1783865941774,
  "partition": 1,
  "offset": 0
}
{
  "topic": "my-redpanda-topic",
  "value": "Testing the rpk producer.",
  "timestamp": 1783865957493.
```

[CLI] deleting a topic

```
redpanda@443d67c64748:/$ rpk topic delete my-redpanda-topic
TOPIC          STATUS
my-redpanda-topic  OK
redpanda@443d67c64748:/$
```